동국대학교
dongguk university

응용수치해석 및 실습 제 2주

권기운

영상을 읽는 모듈

MATPLOTLIB (영상 그리기)

- import matplotlib.pyplot as plt
- import matplotlib.image as mpimg

SKIMAGE

- from skimage.io import imread, imsave, imshow, show
- from skimage import color, viewer, img_as_float, data
- from skimage.transform import SimilarityTransform, warp, swirl, rescale
- from skimage.util import random_noise

PIL

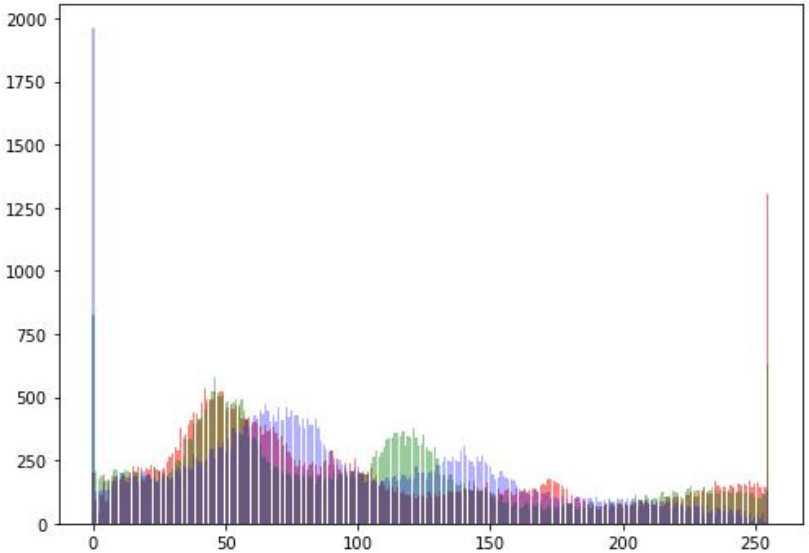
- from PIL import Image, ImageFont, ImageDraw
- from PIL.ImageChops import add, subtract, multiply, difference, screen

SCIPY

- from scipy import misc, ndimage, signal, stats, fftpack as fp

실습 1. 영상 히스토그램

```
from PIL import Image, ImageFont, ImageDraw
import matplotlib.pyplot as plt

im = Image.open("clock.jpg")
plt.figure(figsize=(18,6))
plt.subplot(121), plt.imshow(im)
plt.subplot(122), pl = im.histogram()
plt.bar(range(256),  pl[:256], color='r', alpha = 0.5)
plt.bar(range(256), pl[256:2*256], color='g', alpha = 0.4)
plt.bar(range(256),  pl[2*256:], color='b', alpha = 0.3), plt.show()
```



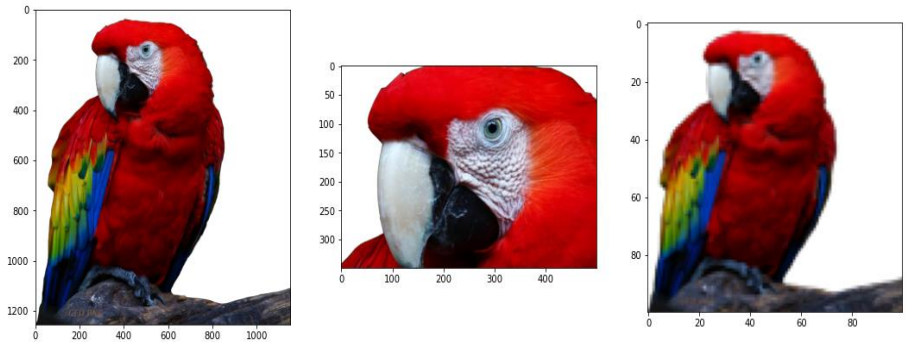
RGB 채널 분리

```
# RGB 채널 분리
ch_r, ch_g, ch_b = im.split()
plt.figure(figsize=(18,6))
plt.subplot(1,3,1), plt.imshow(ch_r, cmap='Reds'), plt.axis('off')
plt.subplot(1,3,2), plt.imshow(ch_g, cmap='Greens'), plt.axis('off')
plt.subplot(1,3,3), plt.imshow(ch_b, cmap='Blues'), plt.axis('off')
plt.tight_layout()  #비교
plt.show()
```

실습 2. 영상 자르기와 크기 조정

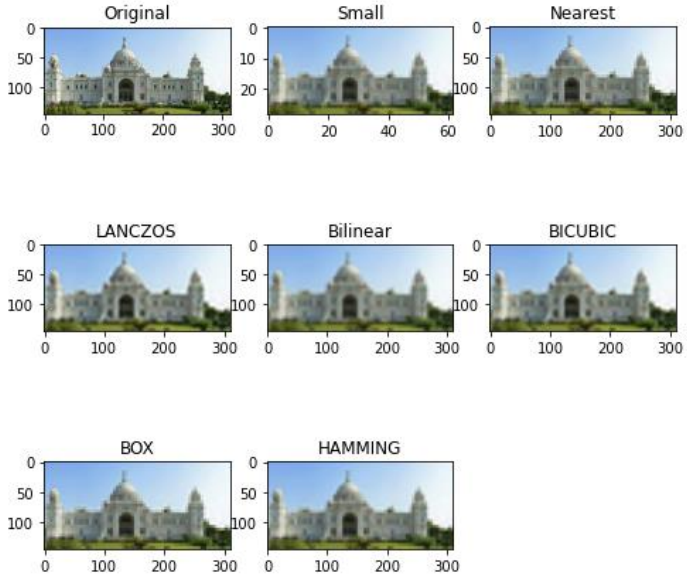
```
from PIL import Image
im = Image.open("parrot.png")
print(im.width, im.height, im.mode, im.format)
>> 1151 1254 RGBA PNG
** Variable Explorer : <PngImageFile @0x1C10E528140> Mode: RGBA
```

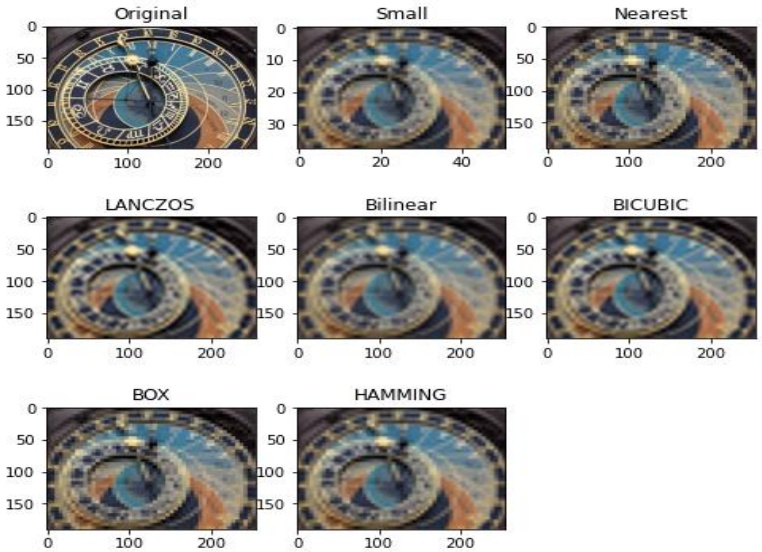
```
im.show()
im_c = im.crop((200,50,700,400))
#left, top, right, bottom
im_c.show()
im2 = im.resize((100,100))
im2.show()

plt.figure(figsize=(18,6))
plt.subplot(1,3,1), plt.imshow(im)
plt.subplot(1,3,2), plt.imshow(im_c)
plt.subplot(1,3,3), plt.imshow(im2)
plt.show()
```



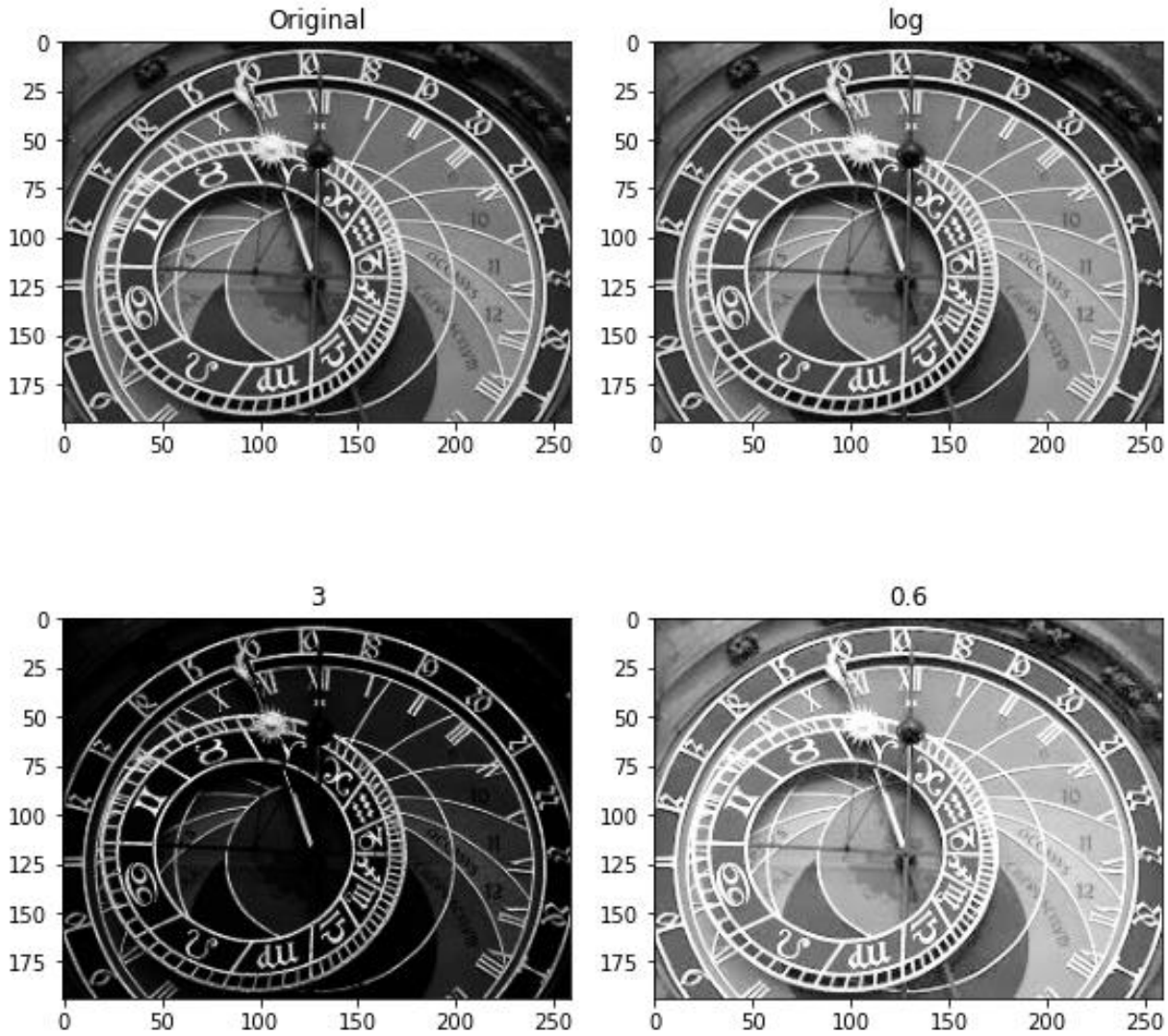
실습 3. 영상 축소와 확대, 다운 샘플링, 업샘플링

```
im = Image.open("clock.jpg")
print(im.width, im.height) #259 194

ims = im.resize((im.width//5,im.height//5),Image.ANTIALIAS)
# 앤티 앨리어싱, 고품질 다운 샘플링 기술
print(ims.width, ims.height)

# Use Image.NEAREST (0), Image.LANCZOS (1),
# Image.BILINEAR (2), Image.BICUBIC (3), Image.BOX (4) or
# Image.HAMMING (5)
im_bilinear = ims.resize((ims.width*5,ims.height*5),Image.BILINEAR)
im_nearest = ims.resize((ims.width*5,ims.height*5),Image.NEAREST)
im_lanczos = ims.resize((ims.width*5,ims.height*5),Image.LANCZOS)
im_bicubic = ims.resize((ims.width*5,ims.height*5),Image.BICUBIC)
im_box = ims.resize((ims.width*5,ims.height*5),Image.BOX)
im_hamming = ims.resize((ims.width*5,ims.height*5),Image.HAMMING)
```



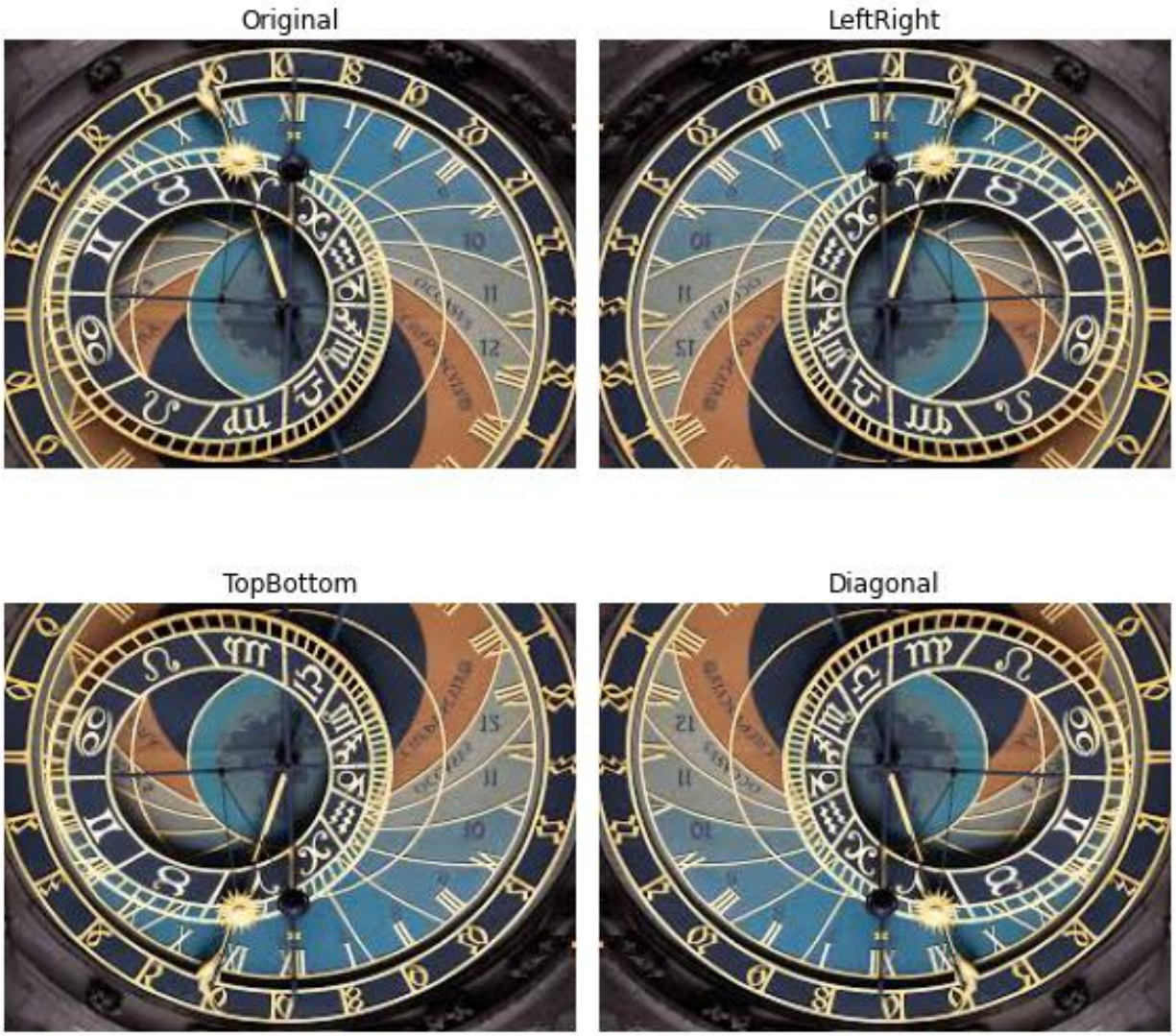


실습 4. 밝기 영상과 로그/파워-로우 변환

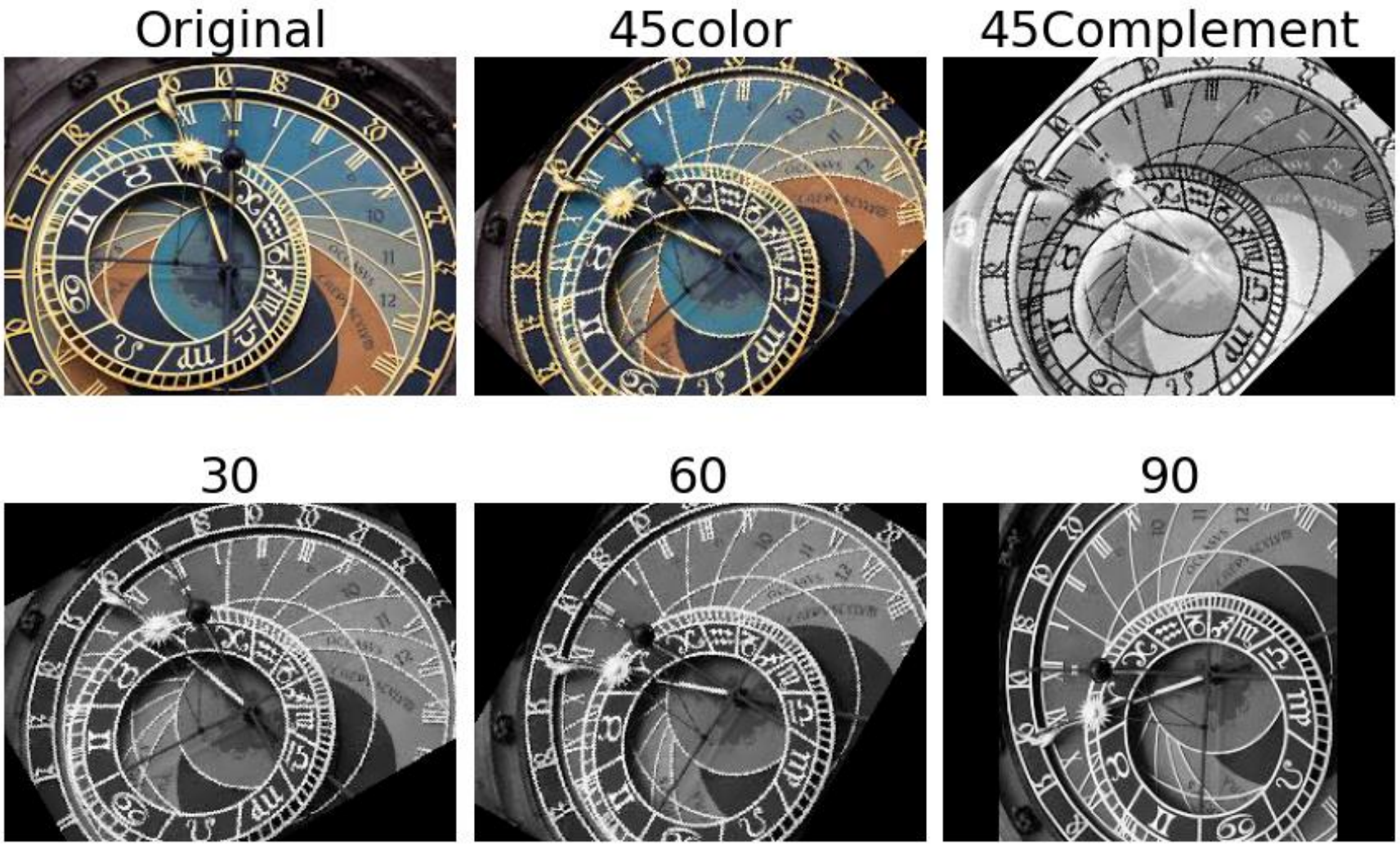


동국대학교
dongguk university

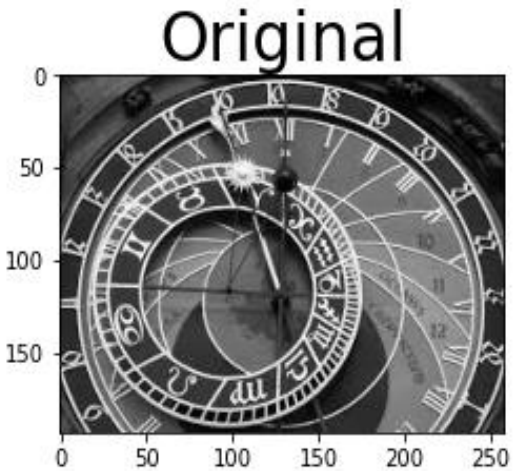
실습 5. 간단한 기하학적 변환

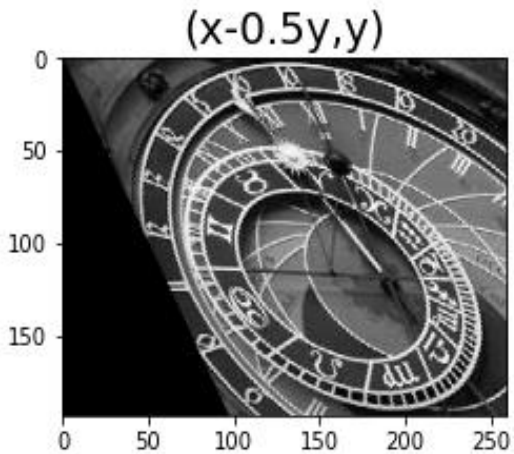


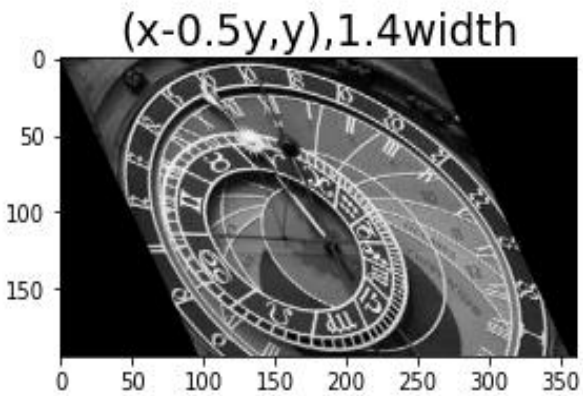
실습 6. 영상 회전

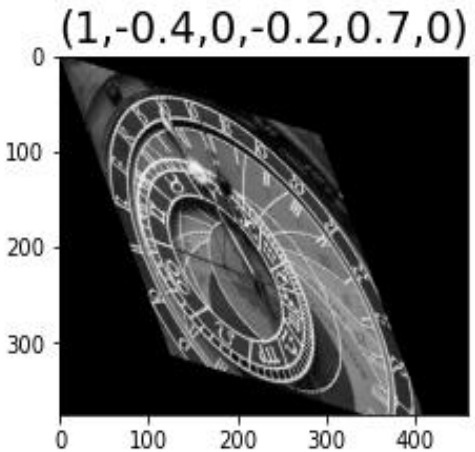


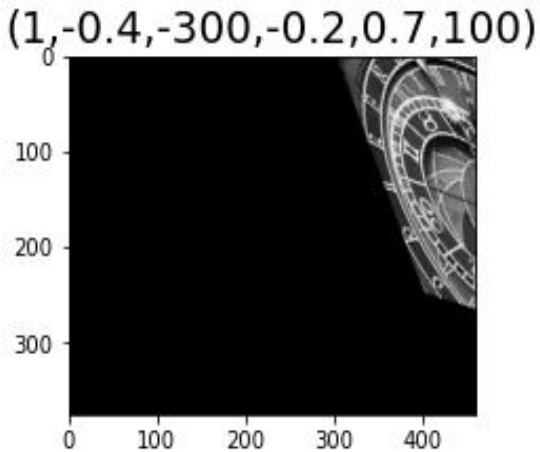
실습 7. 영상 Affine 변환,(x,y)=(ax′+by′+c,dx′+ey′+f)

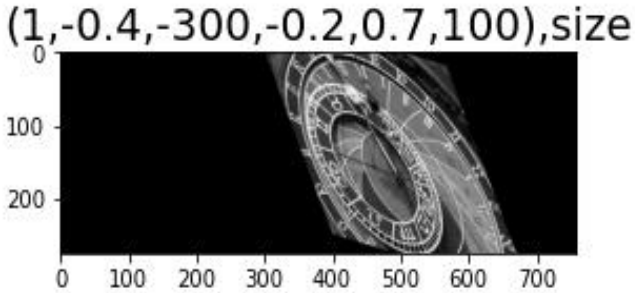




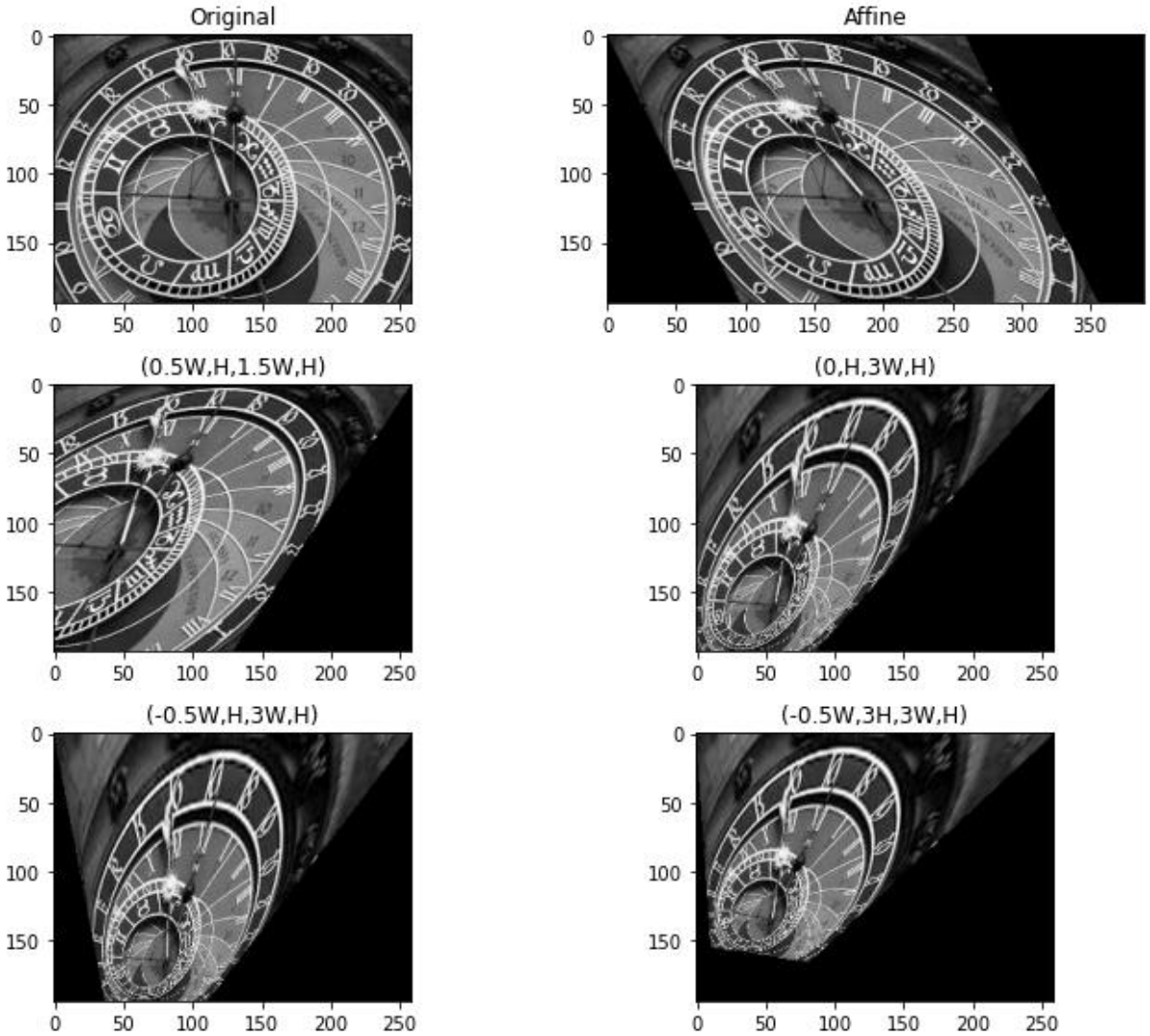




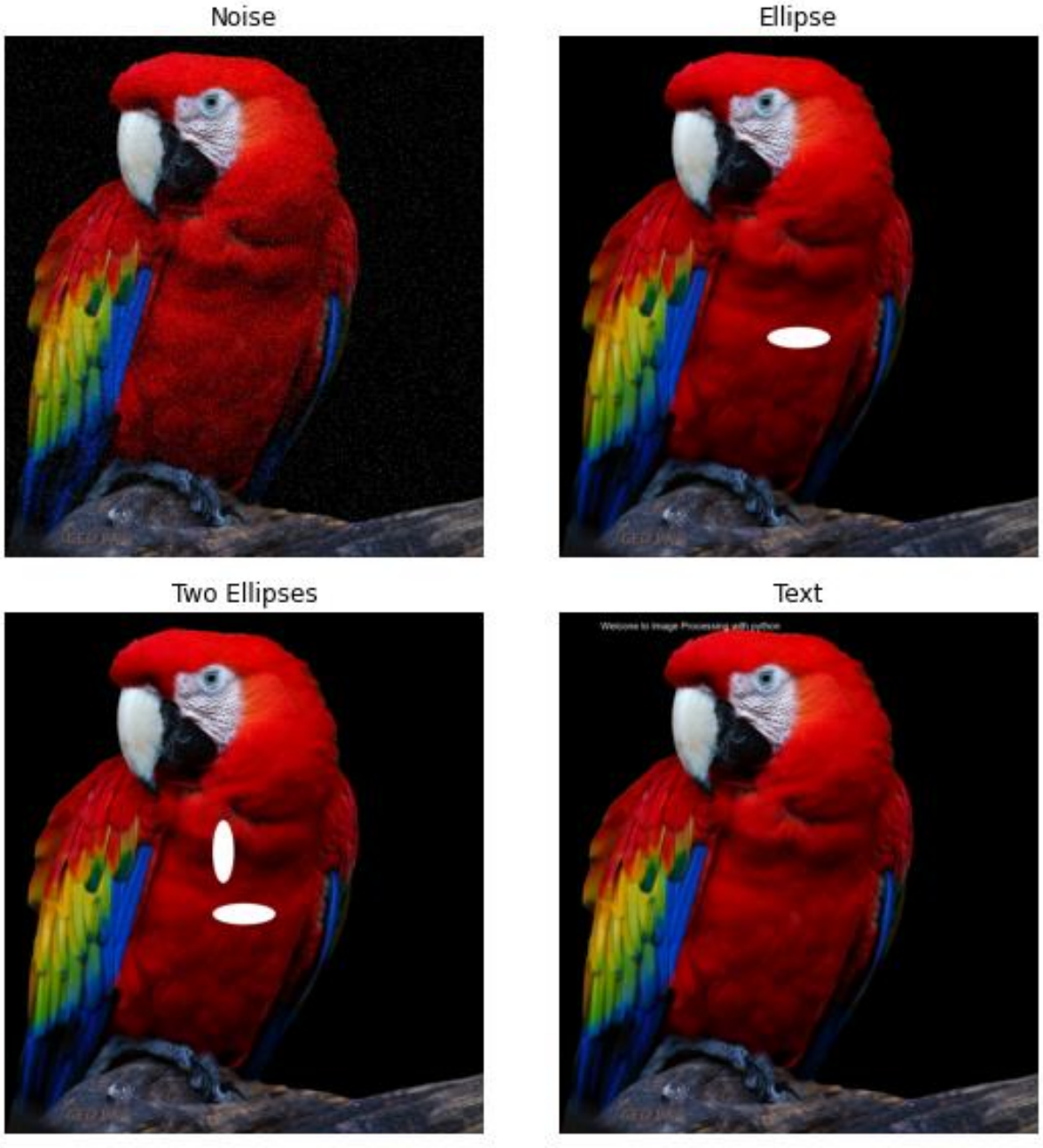




실습 8. 사영변환

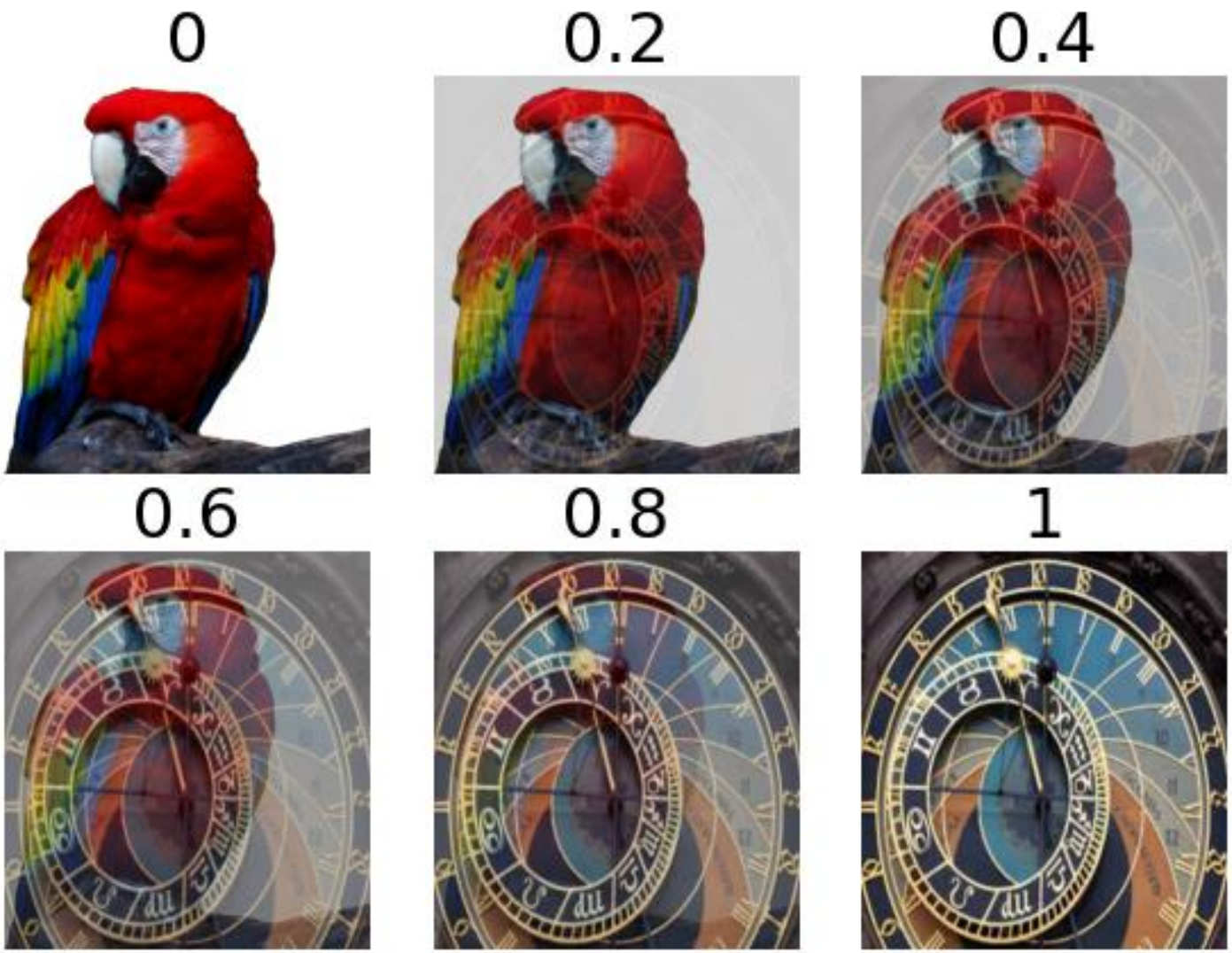


실습 9. 잡음 넣기, 타원/글자 넣기

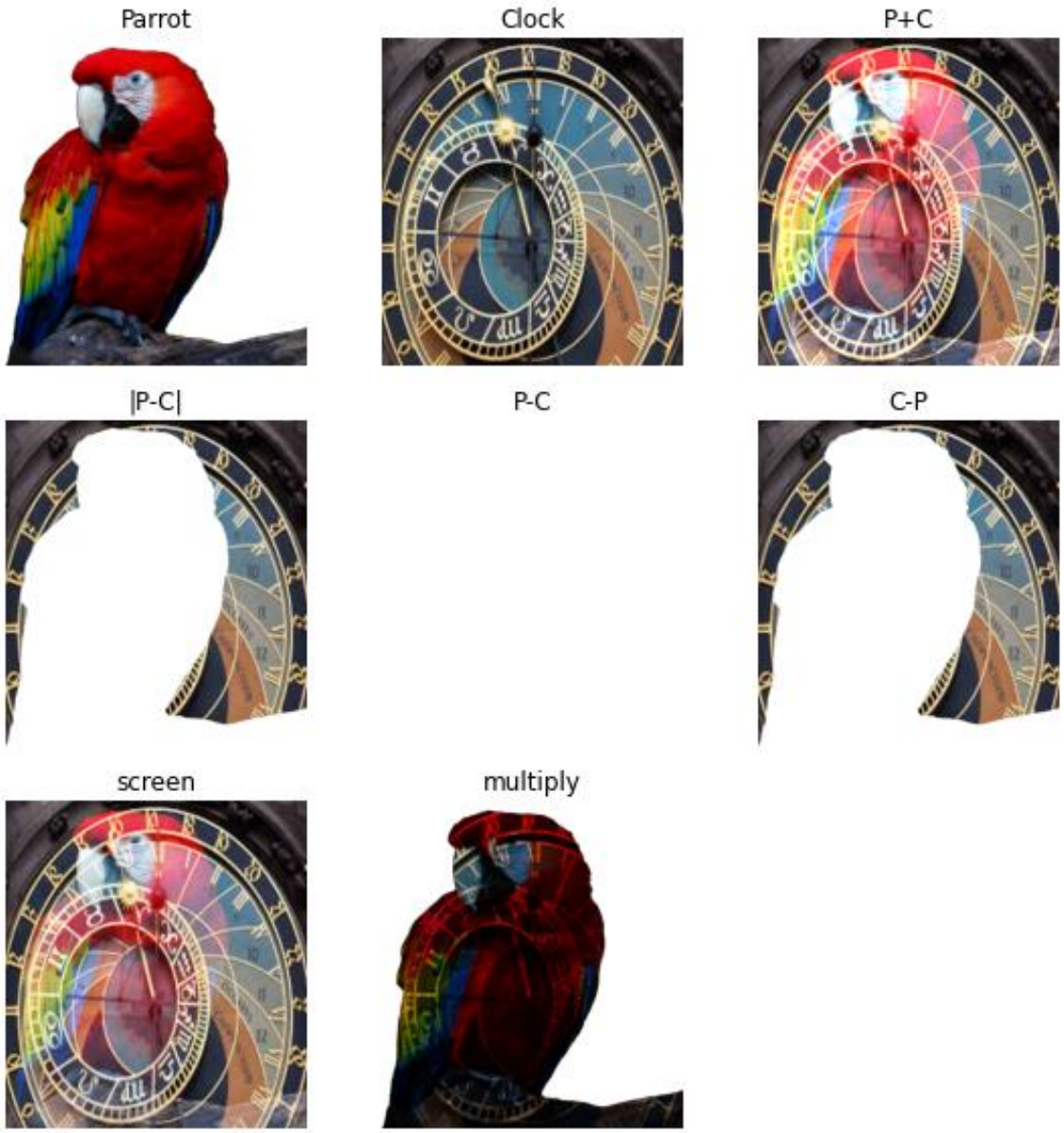


동국대학교
dongguk university

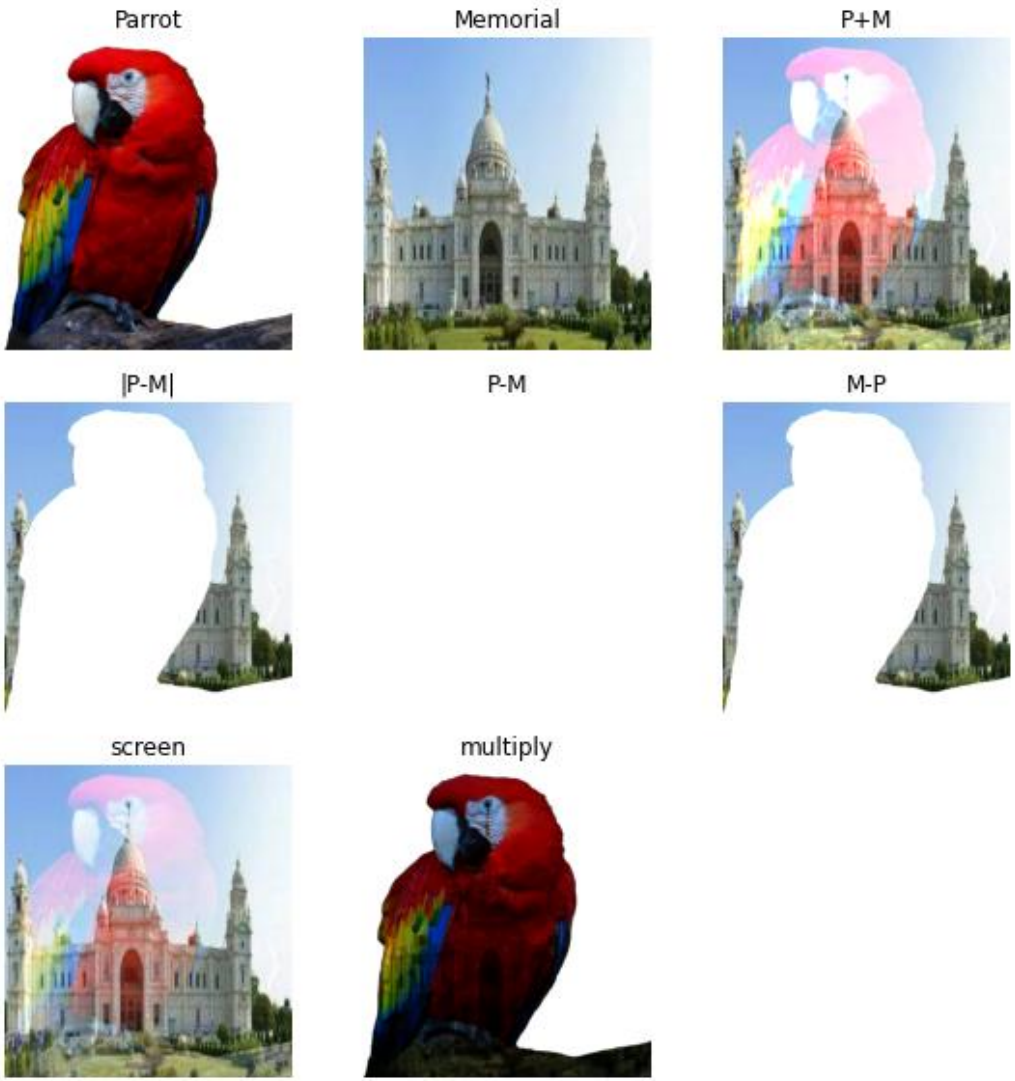
실습 10. 블렌딩



실습 11. 두 영상에 대한 사칙연산, Parrot/Clock



Parrot/Memorial



동국대학교
dongguk university

Clock/Memorial

Clock

Memorial

C+M

|C-M|

C-M

M-C

screen

multiply

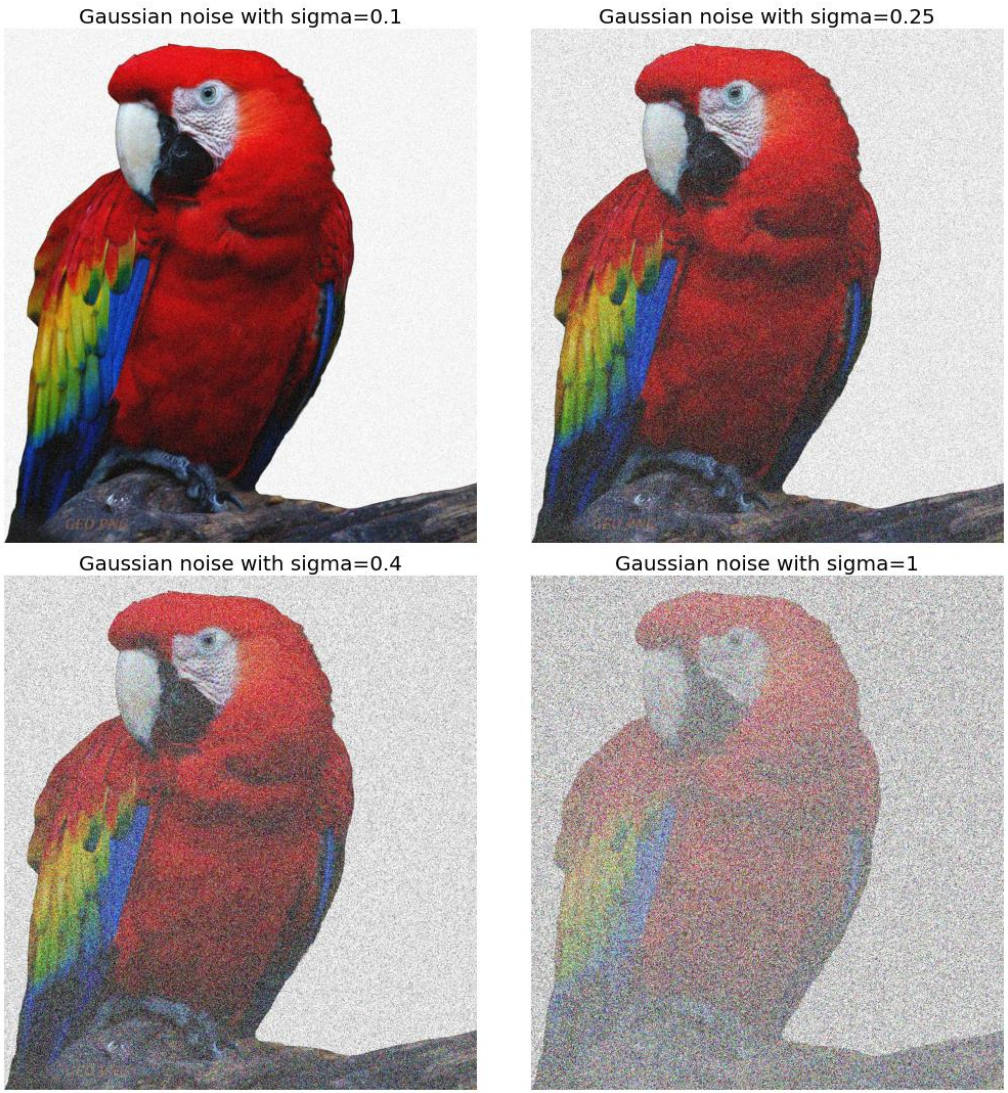
실습 11. 윤곽선

Image Contour Lines

Image Filled Contour

Original Image

실습 12. 가우시안 노이즈



실습 13. Warp 어파인 변환, Swirl 비선형 변환

Warp

Swirl

결론

- 그림은 숫자다.
- 컴퓨터는 숫자 계산을 잘한다.
- 파이썬으로 컴퓨터에 계산을 시킬 수 있다.
- Matplotlib, Scikit-learn 외에도
  PIL 모듈을 사용할 수 있다.